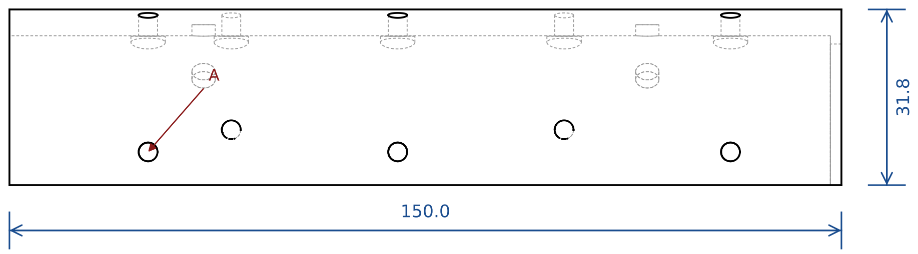
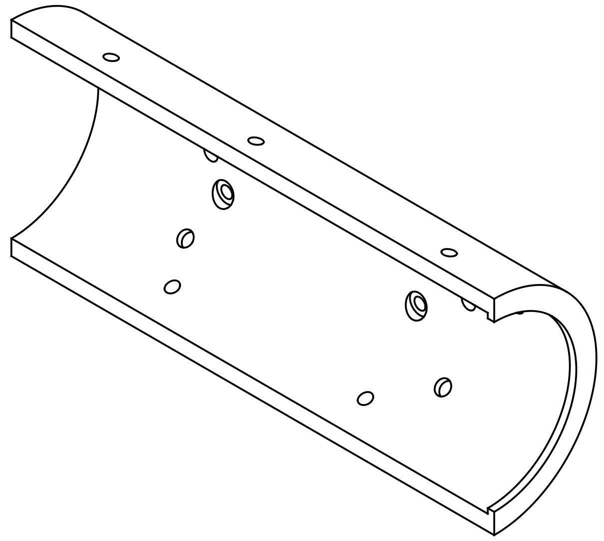
TOP



ISO (not to scale)



NOTES

- Pair of halves from one saw-cut tube, split on the y=0 plane, line-bored together to Ø54.0
- All screw counterbores are cut from the BORE side (heads sit under the liner)
- Rear liner lip: x 148..150, inward to R25.5
- Top rows feed A1 (x 25/70/130); bottom rows feed A3 (x 40/100). Do not mix them up

HOLE / FEATURE SCHEDULE

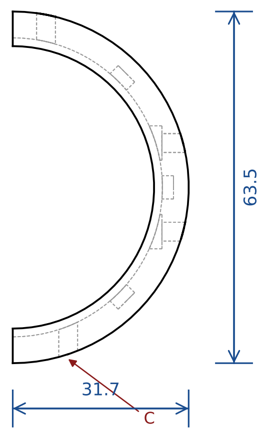
A: 3x Ø3.4 THRU vertical at y=+6 (top crown row), Ø6.2 C'BORE 2.3 deep from the bore @ x 25, 70, 130

B: 3x Ø3.4 THRU horizontal (+y) at z=+8 (top skirt row), Ø6.2 C'BORE from the bore @ x 25, 70, 130

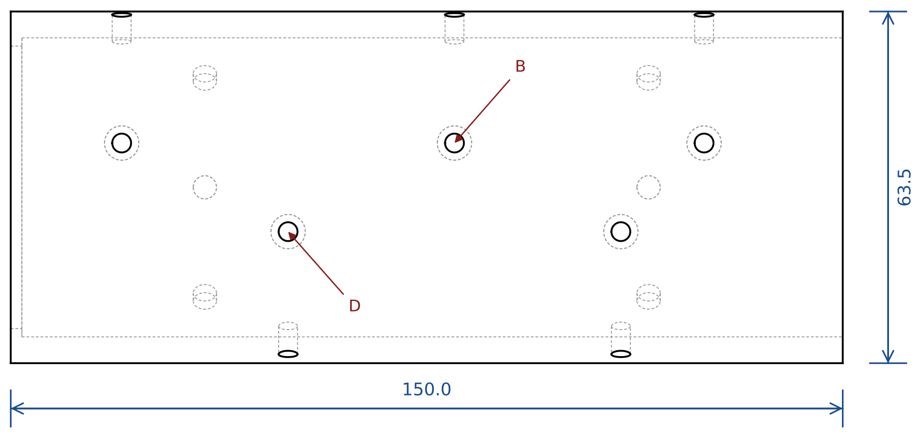
C: 2x Ø3.4 THRU vertical at y=+10 (bottom crown row), Ø6.2 C'BORE from the bore @ x 40, 100

D: 2x Ø3.4 THRU horizontal (+y) at z=-8 (bottom skirt row), Ø6.2 C'BORE from the bore @ x 40, 100

RIGHT



BACK

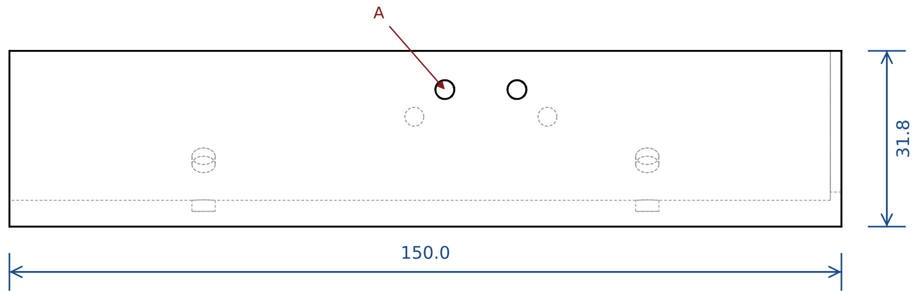


GENERAL

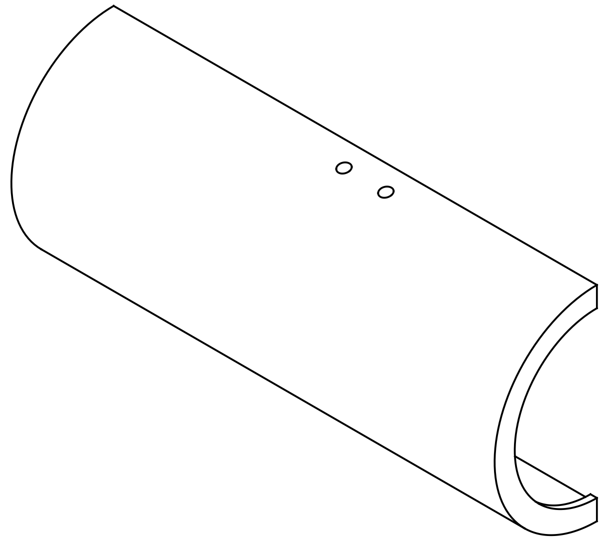
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061-T6 tube 2.5" OD x 3/16" wall, or PETG (stage 1)	SCALE (ortho views) 0.73 : 1	DATE 2026-09-06	
C1 - chassis half (carries the boxes) file: cnc-design/step/C1_chassis_half.step	QTY 1	BOUNDING BOX x / y / z 150.0 / 31.8 / 63.5 (x -0.0..150.0, y -0.0..31.8, z -31.8..31.8)	SHEET 1 / 12	3rd angle A3 landscape

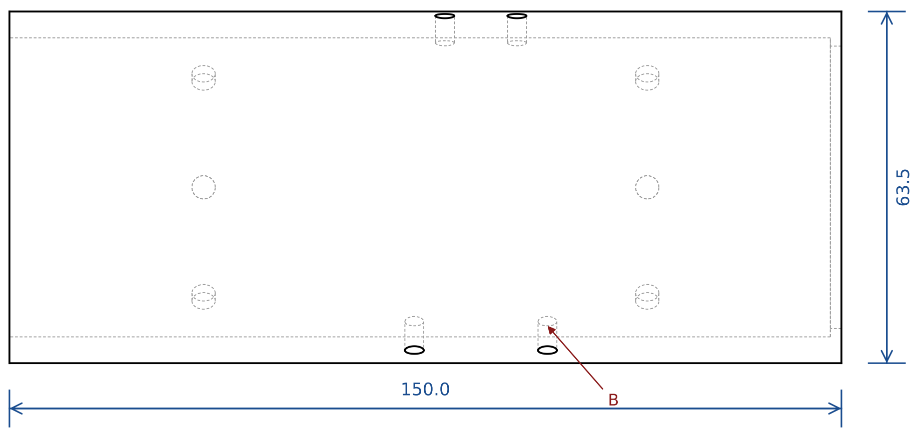
TOP



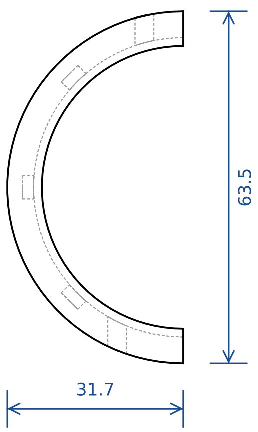
ISO (not to scale)



FRONT



RIGHT



NOTES

- Mate of C1 (same tube, same line-bore)
- Counterbores from the BORE side
- Rear liner lip: x 148..150, inward to R25.5

HOLE / FEATURE SCHEDULE

A: 2x Ø3.4 THRU vertical at y=-7 (closure block), Ø6.2 C'BORE from the bore @ x 78.5, 91.5

B: 2x Ø3.4 THRU vertical at y=-11.9 through the BOTTOM wall (hinge block), Ø6.2 C'BORE from the bore @ x 73, 97

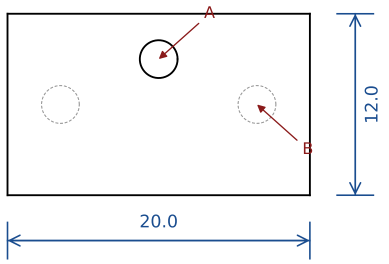
S: 6x Ø4.2 +0.1/-0 x 2 BLIND radial, from the bore (liner press-fit studs) @ x 35, 115 at -45/0/+45 deg from the mid-arc (6 per half)

GENERAL

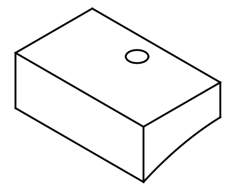
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061-T6 tube 2.5" OD x 3/16" wall, or PETG (stage 1)	SCALE (ortho views) 0.73 : 1	DATE 2026-09-06
C2 - clamp half (hinged) file: cnc-design/step/C2_clamp_half.step	QTY 1	BOUNDING BOX x / y / z 150.0 / 31.8 / 63.5 (x -0.0..150.0, y -31.8..0.0, z -31.8..31.8)	SHEET 2 / 12 3rd angle A3 landscape

TOP



ISO (not to scale)



NOTES

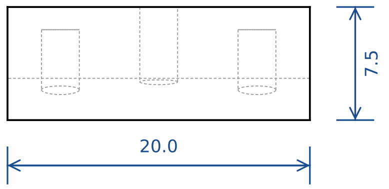
- Underside is a saddle R31.8 - sits on C2's OD
- Top face is flat and CONTACTS the A1 pocket roof under the closure screw
- The closure screw comes down the latch bore: M3x8 low-head

HOLE / FEATURE SCHEDULE

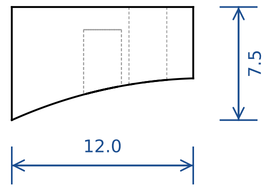
A: M3 closure-screw thread THRU, M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) @ (85, -4)

B: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG), from the saddle side, 6 deep @ x 78.5, 91.5 at y=-7

FRONT



RIGHT

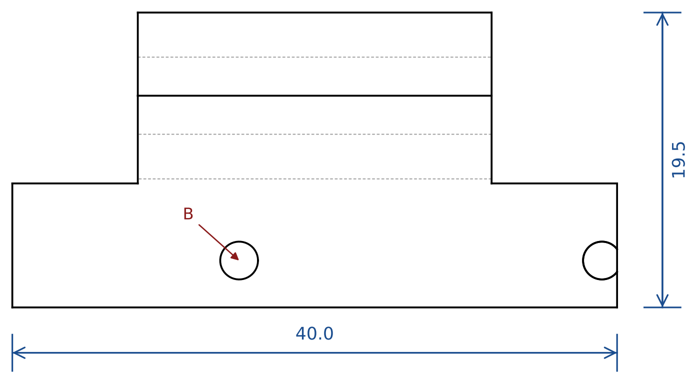


GENERAL

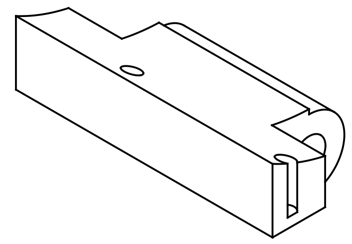
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL steel or 6061 (PETG stage 1)	SCALE (ortho views) 2.00 : 1	DATE 2026-09-06	
B - closure block (on C2, under A1) file: cnc-design/step/closure_block.step	QTY 1	BOUNDING BOX x / y / z 20.0 / 12.0 / 7.5 (x 75.0..95.0, y -13.0..-1.0, z 29.0..36.5)	SHEET 3 / 12	3rd angle A3 landscape

TOP



ISO (not to scale)



NOTES

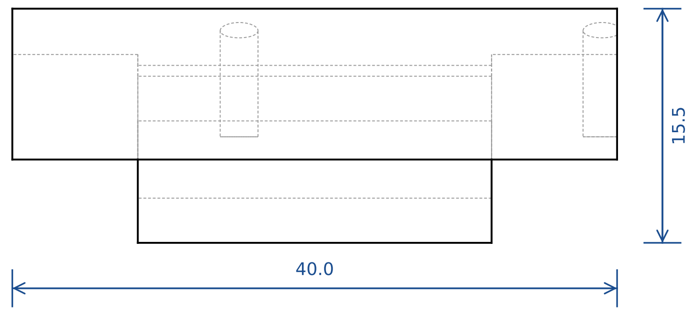
- Body rides C2's OD (saddle R31.8); lug is round about the pin
- Body bottom = pin height by design (swing clearance) - do not extend it
- Lug spans x 66.3..89.7 (0.3 end float between A3's lugs)

HOLE / FEATURE SCHEDULE

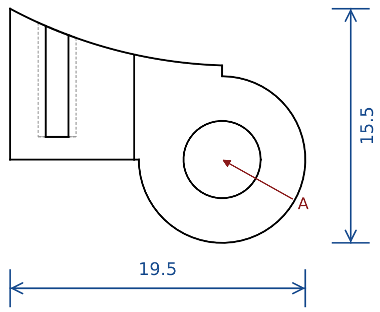
A: Ø5.1 pin bore THRU the lug, axis at (y -1, z -38) @ along x

B: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG), from the saddle side @ x 73, 97 at y=-11.9

FRONT



RIGHT

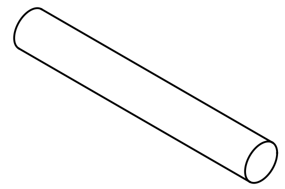


GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061-T6 (PETG stage 1)	SCALE (ortho views) 2.00 : 1	DATE 2026-09-06	
H - hinge block (on C2, carries C2's lug) file: cnc-design/step/hinge_block.step	QTY 1	BOUNDING BOX x / y / z 40.0 / 19.5 / 15.5 (x 58.0..98.0, y -15.0..4.5, z -43.5..-28.0)	SHEET 4 / 12	3rd angle A3 landscape

ISO (not to scale)



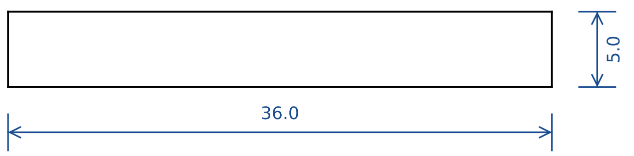
NOTES

- Standard Ø5 x 36 dowel (cut a 40 to length if needed); ends chamfered
- Assembly: enter through A3 lug 1 (x 58 face) to the blind end, then press a Ø5 x 2 plug flush behind it

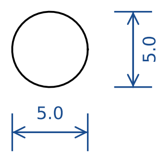
HOLE / FEATURE SCHEDULE

A: Ø5 h6 x 36

FRONT



RIGHT

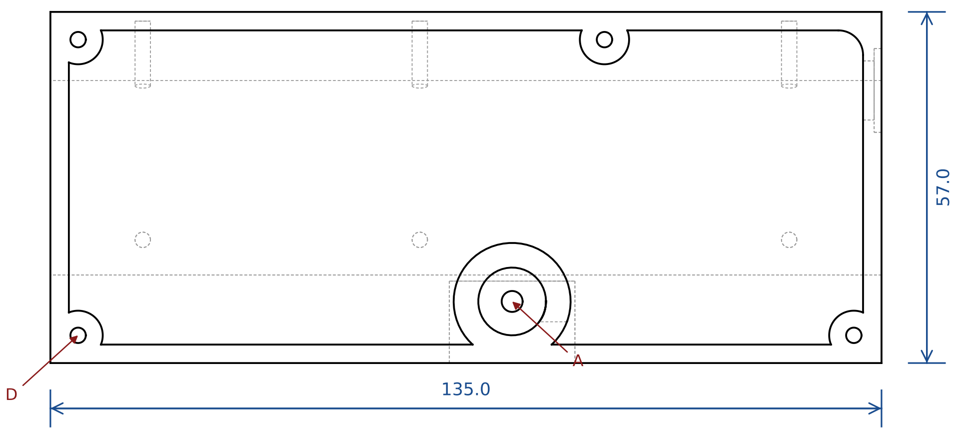


GENERAL

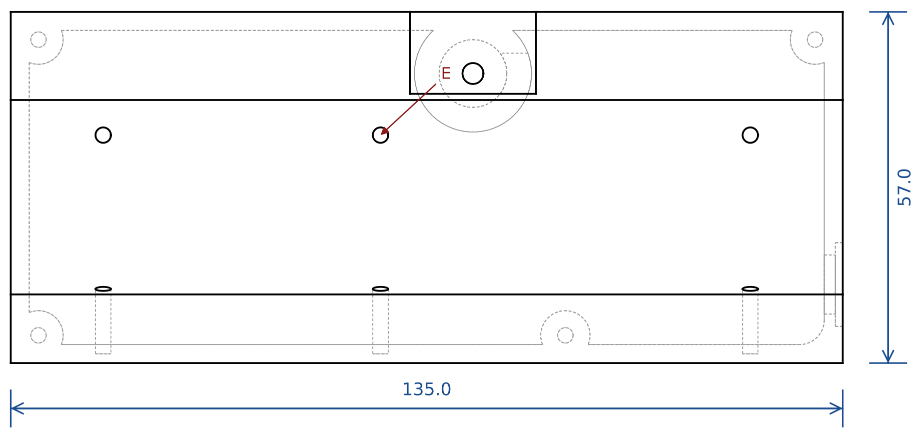
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL steel dowel Ø5 h6	SCALE (ortho views) 2.00 : 1	DATE 2026-09-06	
hinge pin file: cnc-design/step/hinge_pin.step	QTY 1	BOUNDING BOX x / y / z 36.0 / 5.0 / 5.0 (x 60.0..96.0, y -3.5..1.5, z -40.5..-35.5)	SHEET 5 / 12	3rd angle A3 landscape

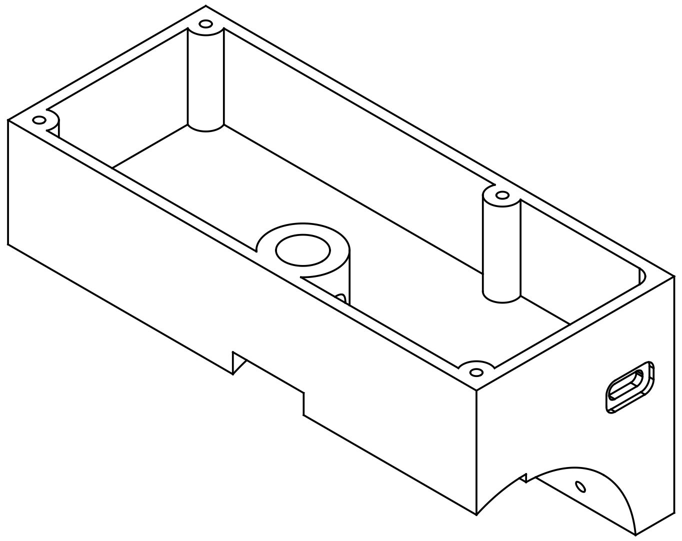
TOP



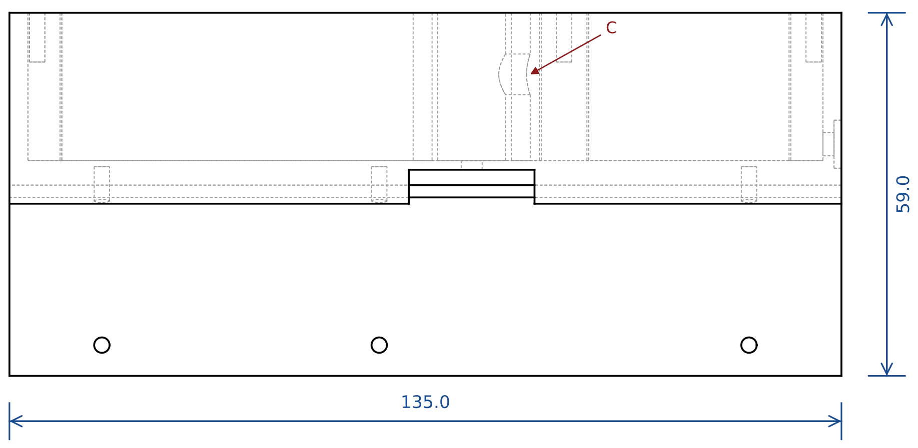
BOTTOM



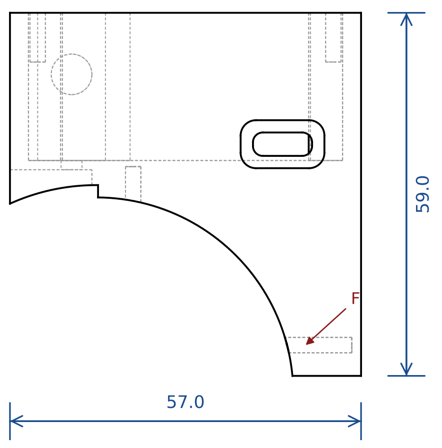
ISO (not to scale)



FRONT



RIGHT



NOTES

- Underside: saddle R32.0 on the C1 side, relieved to R34.0 over C2 (y<0.3) - C2's rim arcs outward as it swings
- Interior pocket 129 x 51 x 24 deep from z=38, R6 corners
- Closure-block pocket under the -y overhang: x 74.8..95.2, open toward -y, roof at z=36.5
- Latch boss Ø19 rises from the floor around the receiver bore

HOLE / FEATURE SCHEDULE

A: Ø11 receiver bore from the top down to the floor (z 38) @ (85, -4)

B: Ø3.4 closure-screw clearance THRU the floor (1.5 thick) under A @ same axis

C: Ø6.6 plunger channel, horizontal along +x from the bore, axis z=52 @ (x 85.., y -4)

D: 4x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) lid screws, 8 deep from the top @ (14.5, -9.5), (14.5, 38.5), (140.5, -9.5), (100, 38.5)

E: 3x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) vertical from the saddle at y=+6 (crown row), to z=37 @ x 25, 70, 130

F: 3x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) horizontal (+y) at z=+8 into the skirt (skirt row) @ x 25, 70, 130

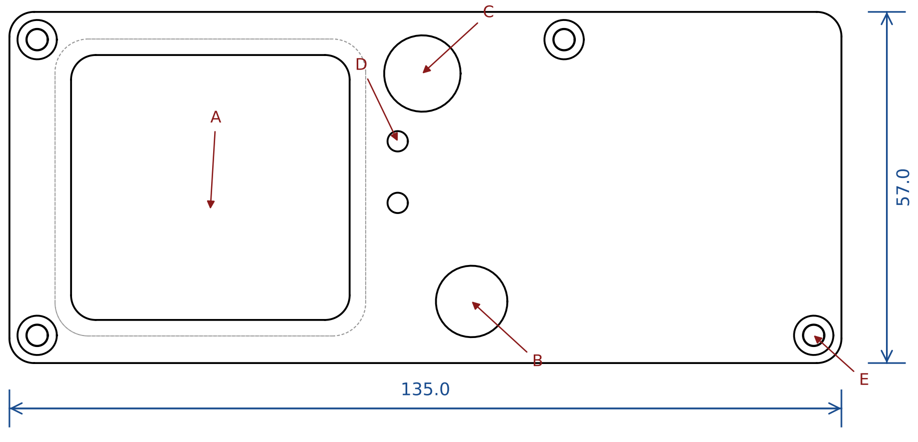
G: USB-C slot 9.6 x 3.8 THRU the +x end wall, R1.6, + plug recess 13.6 x 7.8 x 2 deep outside @ y 30.25, z 40.65

GENERAL

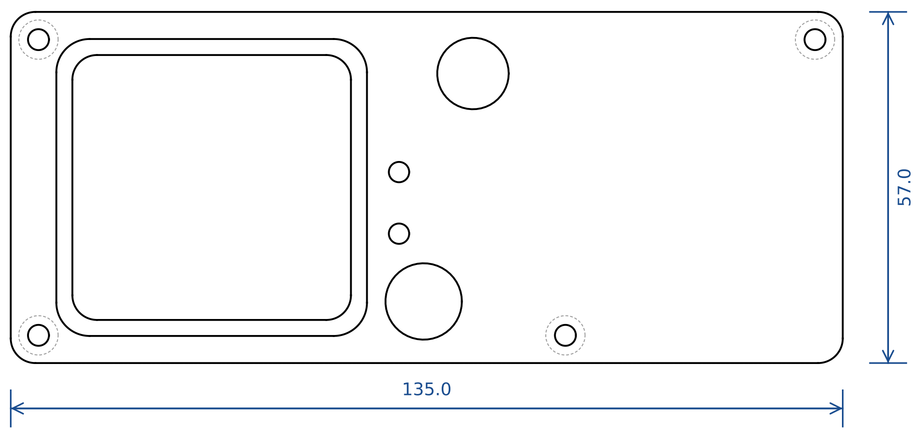
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061-T6 billet (PETG stage 1)	SCALE (ortho views) 0.81 : 1	DATE 2026-09-06	
A1 - top box (latch + electronics) file: cnc-design/step/A1_top_box.step	QTY 1	BOUNDING BOX x / y / z 135.0 / 57.0 / 59.0 (x 10.0..145.0, y -14.0..43.0, z 3.0..62.0)	SHEET 6 / 12	3rd angle A3 landscape

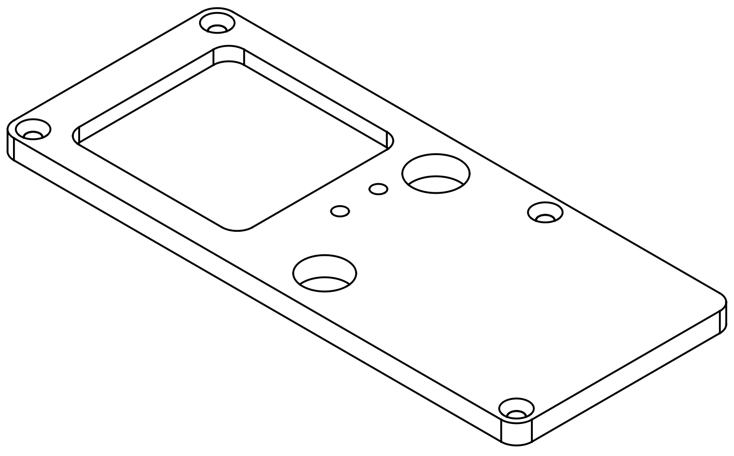
TOP



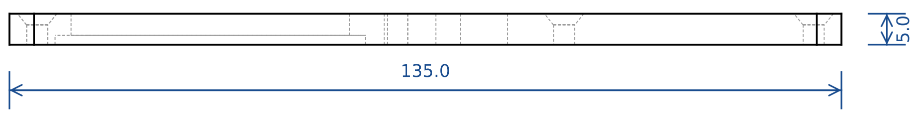
BOTTOM



ISO (not to scale)



FRONT



NOTES

- RF window 45.2 x 43 THRU, R4 corners; underside recess 50.4 x 48.2 x 1.5 deep for the insert flange
- Lid screws countersunk 90deg from the top for M3 flat head

HOLE / FEATURE SCHEDULE

A: window 45.2 x 43 THRU @ centre (42.6, 14.5)

B: Ø11.6 THRU (latch bore pass-through) @ (85, -4)

C: Ø12.4 THRU (sealed button) @ (77, 33)

D: 2x Ø3.3 THRU (LEDs) @ (73, 12) (73, 22)

E: 4x Ø3.4 THRU, 90deg C'SINK Ø6.4 from the top @ (14.5, -9.5), (14.5, 38.5), (140.5, -9.5), (100, 38.5)

GENERAL

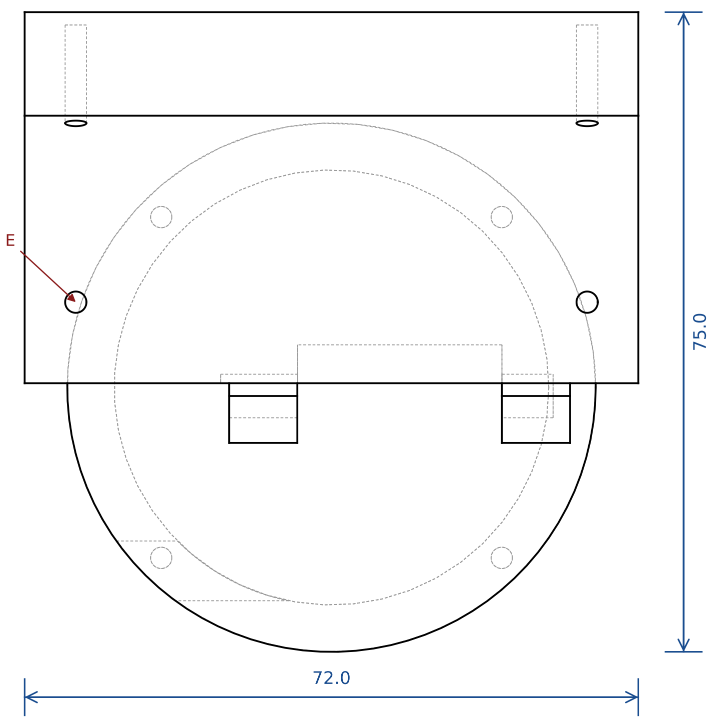
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.

Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.

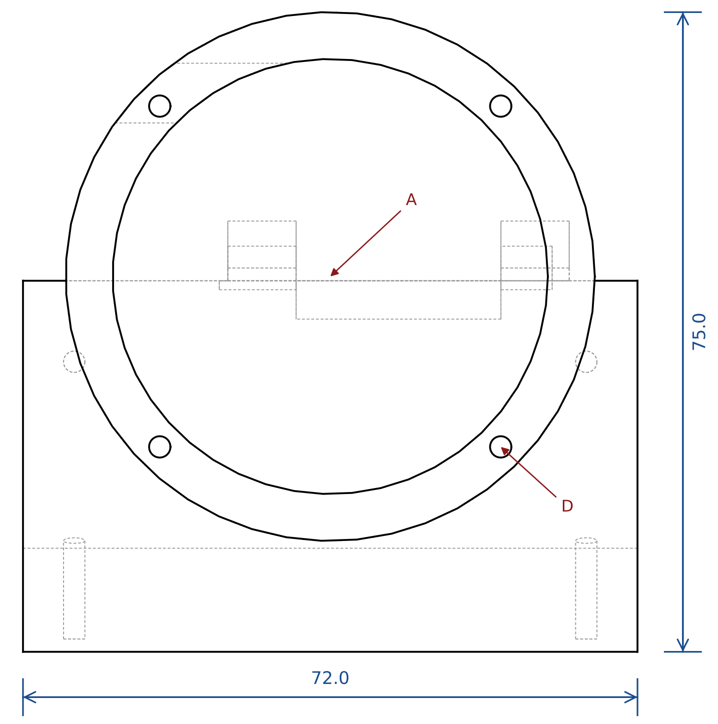
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061 plate 5 mm (PETG stage 1)	SCALE (ortho views) 0.81 : 1	DATE 2026-09-06	
A2 - lid plate file: cnc-design/step/A2_lid.step	QTY 1	BOUNDING BOX x / y / z 135.0 / 57.0 / 5.0 (x 10.0..145.0, y -14.0..43.0, z 62.0..67.0)	SHEET 7 / 12	3rd angle A3 landscape

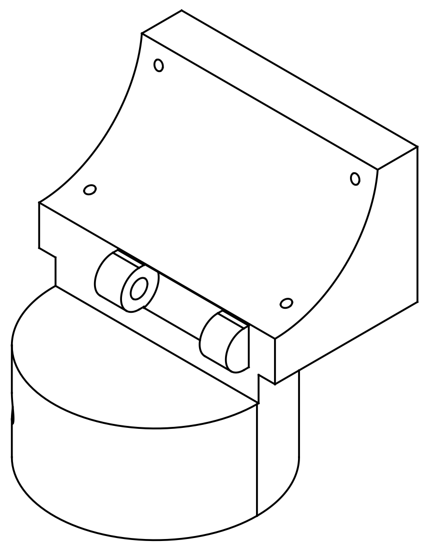
TOP



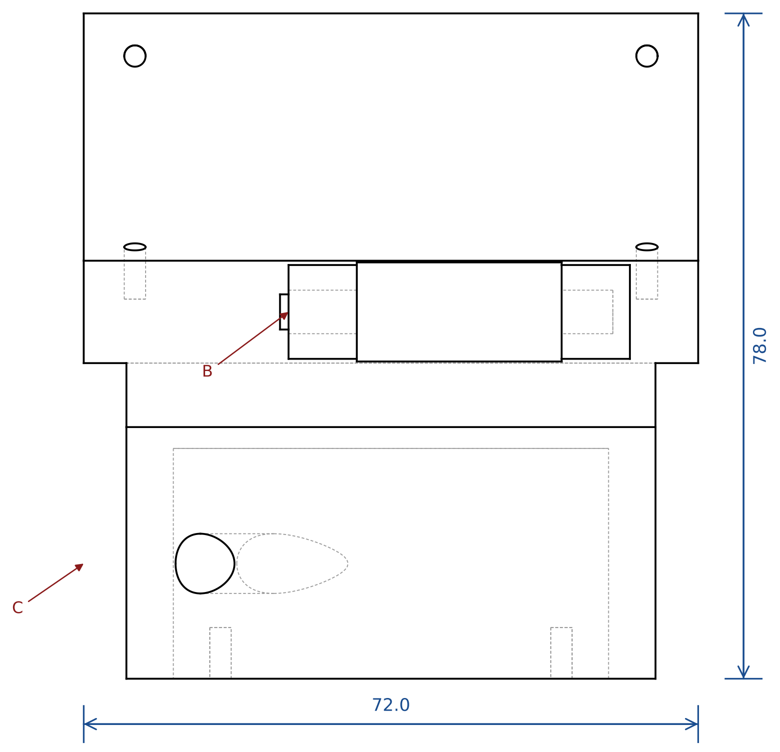
BOTTOM



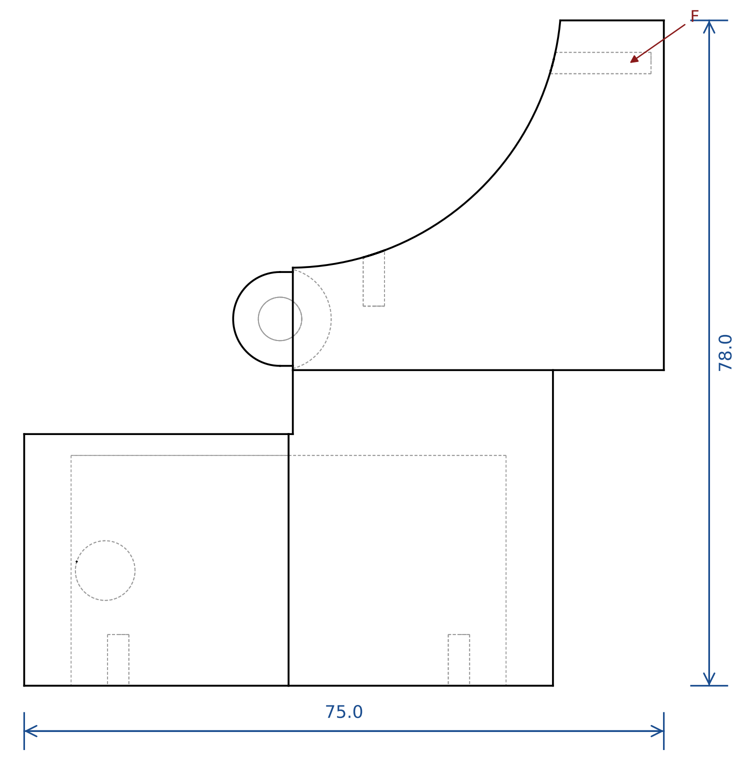
ISO (not to scale)



FRONT



RIGHT



NOTES

- Cradle x 34..106, y 0.5..44, saddle R32.0 on top; skirt rib y 32..44 up to z=-3
- Puck Ø62 centred under the tube (x 70, y 0); flat top at z=-51.5 on the C2 side = hinge STOP
- Two hinge lugs Ø11 about the pin axis (y -1, z -38) at x 58..66 and 90..98; Ø12 clearance between them for C2's lug
- Spool cartridge + power spring load from below; A4 closes the pocket

HOLE / FEATURE SCHEDULE

A: Ø51 spool pocket from the bottom face up to z=-54 (depth 27) @ axis (70, 0)

B: Ø5.1 pin bore along x from lug 1's outer face (x 58), BLIND, ends at x 96 @ axis (y -1, z -38)

C: Ø7 cable exit along x through the puck wall (fit a steel bushing) @ y -21.5, z -67.5

D: 4x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) cover screws, 6 deep from the bottom face, in the puck wall @ PCD 56.5 at 45/135/225/315deg

E: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) vertical from the saddle at y=10 (crown row) @ x 40, 100

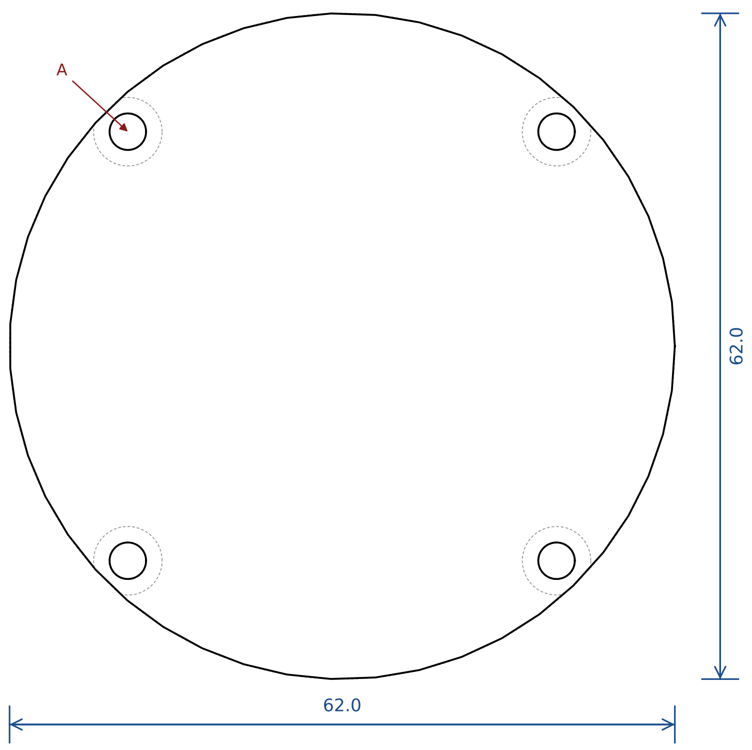
F: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) horizontal (+y) at z=-8 into the skirt rib (skirt row) @ x 40, 100

GENERAL

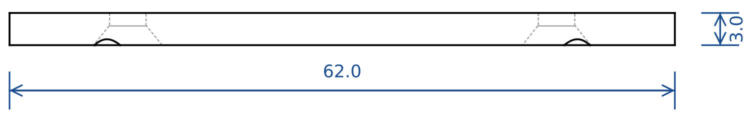
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061-T6 billet (PETG stage 1)	SCALE (ortho views) 1.13 : 1	DATE 2026-09-06	
A3 - spool puck + cradle (hinge knuckles) file: cnc-design/step/A3_bottom_box.step	QTY 1	BOUNDING BOX x / y / z 72.0 / 75.0 / 78.0 (x 34.0..106.0, y -31.0..44.0, z -81.0..-3.0)	SHEET 8 / 12	3rd angle A3 landscape

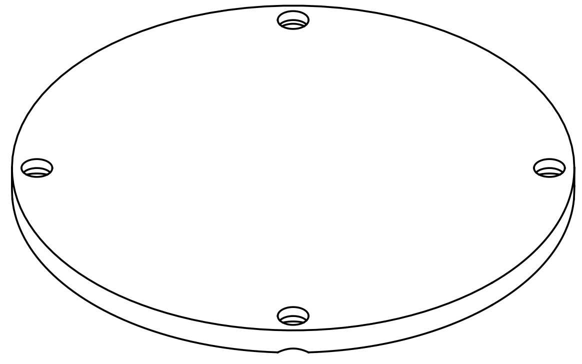
TOP



FRONT



ISO (not to scale)



NOTES

- Ø62 disc, 3 thick; gasket under it
- Screws countersunk 90deg from the outside for M3 flat head

HOLE / FEATURE SCHEDULE

A: 4x Ø3.4 THRU, 90deg C'SINK Ø6.4 @ PCD 56.5 at
45/135/225/315deg

GENERAL

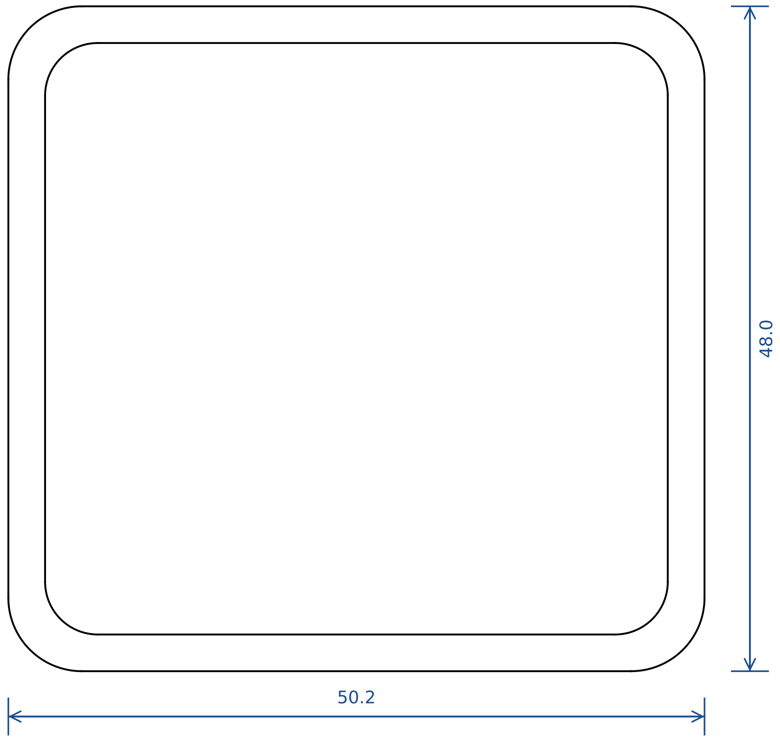
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.

Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.

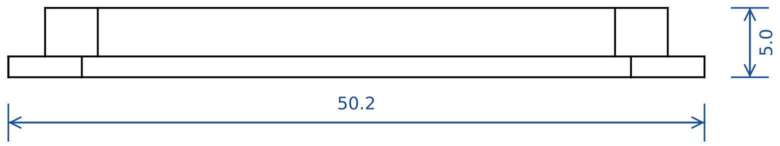
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL 6061 plate 3 mm (PETG stage 1)	SCALE (ortho views) 1.42 : 1	DATE 2026-09-06	
A4 - puck cover file: cnc-design/step/A4_cover_plate.step	QTY 1	BOUNDING BOX x / y / z 62.0 / 62.0 / 3.0 (x 39.0..101.0, y -31.0..31.0, z -84.0..-81.0)	SHEET 9 / 12	3rd angle A3 landscape

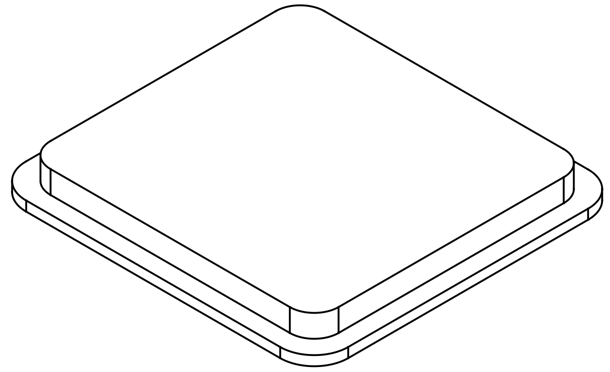
TOP



FRONT



ISO (not to scale)



NOTES

- Body fills the lid cutout flush with the lid top; flange sits in the lid's underside recess
- Opaque by choice - RF does not care; keep it non-metallic

HOLE / FEATURE SCHEDULE

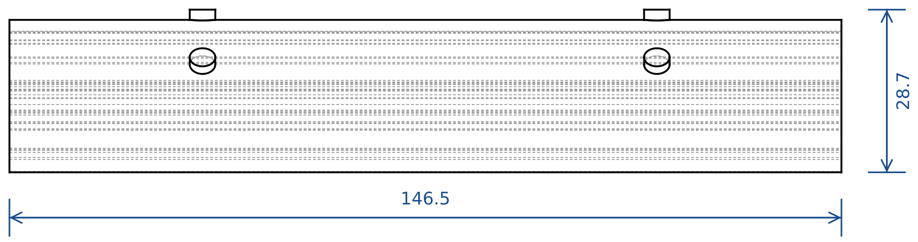
A: body 44.9 x 42.7 x 3.5, flange 50.2 x 48 x 1.5

GENERAL

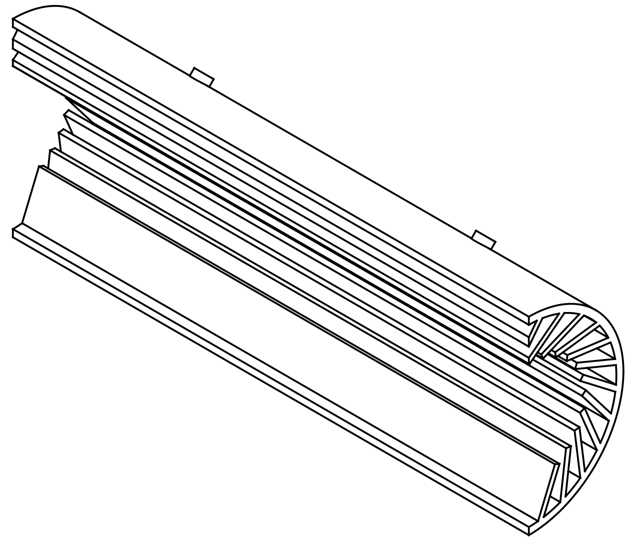
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL printed PETG/ASA or polycarbonate	SCALE (ortho views) 1.83 : 1	DATE 2026-09-06	
A5 - RF window insert (opaque) file: cnc-design/step/A5_window_insert.step	QTY 1	BOUNDING BOX x / y / z 50.2 / 48.0 / 5.0 (x 17.5..67.7, y -9.5..38.5, z 62.0..67.0)	SHEET 10 / 12	3rd angle A3 landscape

TOP



ISO (not to scale)



NOTES

• Base ring R24.85..26.85 with 24 fins (1.2 wide x 4 tall) - fins bend to grip Ø32..46 down tubes

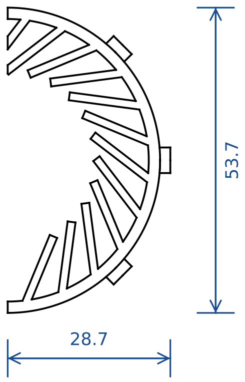
• Left half is the mirror (liner_left)

HOLE / FEATURE SCHEDULE

A: 24 fins on 15deg pitch, 1.4 x 12, leaning 30deg

B: 6x press-fit studs Ø4.5 ±0.1 x 1.8 on the outside; mate: Ø4.2 +0.1/-0 x 2 blind holes in the tube wall (0.3 interference) @ x 35, 115 at -45/0/+45 deg from the mid-arc (6 per half)

RIGHT



GENERAL

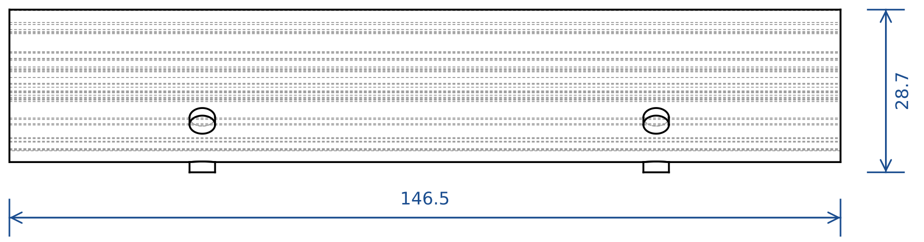
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.

Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.

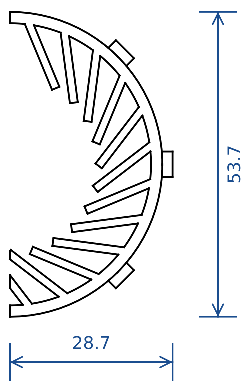
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL TPU 95A, printed on its seam face	SCALE (ortho views) 0.75 : 1	DATE 2026-09-06	
liner (right half, C1 side) file: cnc-design/step/liner_right.step	QTY 1	BOUNDING BOX x / y / z 146.5 / 28.7 / 53.7 (x 1.0..147.5, y -0.0..28.7, z -26.9..26.9)	SHEET 11 / 12	3rd angle A3 landscape

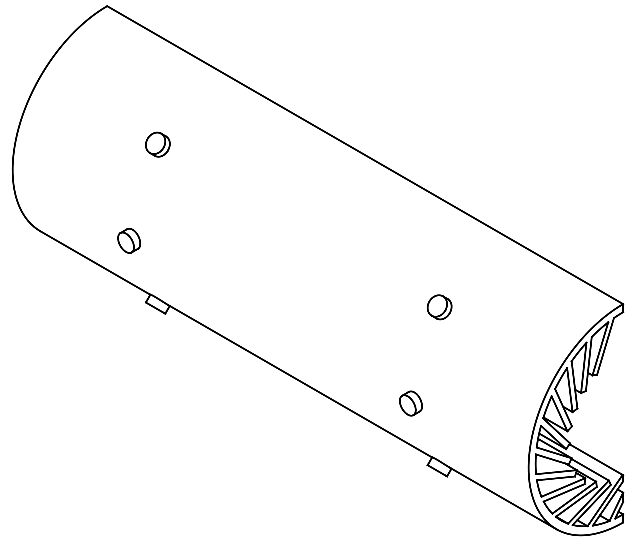
TOP



LEFT



ISO (not to scale)



NOTES

- Mirror of liner_right

HOLE / FEATURE SCHEDULE

A: 24 fins on 15deg pitch, 1.4 x 12, leaning 30deg

B: 6x press-fit studs Ø4.5 x 1.8 - see liner_right

GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.

Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.

Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING cad/cnc_casing_cq.py rev 3b	MATERIAL TPU 95A, printed on its seam face	SCALE (ortho views) 0.75 : 1	DATE 2026-09-06	
liner (left half, C2 side) file: cnc-design/step/liner_left.step	QTY 1	BOUNDING BOX x / y / z 146.5 / 28.7 / 53.7 (x 1.0..147.5, y -28.7..0.0, z -26.9..26.9)	SHEET 12 / 12	3rd angle A3 landscape